Urbanism Quality, Housing Prices, and Effective Housing Supply:

Evidence from Israeli Cities

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Abstract

This paper links observable urbanism metrics—street-network density, junction density, amenity density, and composite walkability—to residential apartment prices across 21 Israeli cities, using administrative transaction data for 2018–2023 and street-network/amenity data from OpenStreetMap. We find that street density (km/km²) and amenity density (POIs/km²) are the strongest cross-city predictors of median apartment prices, with Pearson correlations of 0.63 and 0.64 respectively (both $p < 0.01$), while a composite walkability index carries a Pearson correlation of 0.44 ($p < 0.05$). Dead-end ratio and circuity are not significantly associated with prices at the city level. We then embed these findings in a broader theoretical framework, drawing on the hedonic price literature and on spatial equilibrium models of housing supply. We argue that the observed capitalization of neighborhood walkability and amenity access is precisely the micro-level evidence needed to construct a quality-adjusted or 'effective' housing supply measure—analogous to efficiency-weighted labor in labor economics—in which each dwelling unit contributes to the aggregate stock in proportion to its hedonic-predicted value. Accounting for this quality margin implies that the Israeli housing shortage is more severe than raw unit counts suggest, because a disproportionate share of the existing stock is located in low-connectivity, low-amenity urban environments that provide fewer 'housing services' per physical unit.

JEL Codes: R21, R31, R41, R52, D40   Keywords: hedonic prices, walkability, street-network connectivity, amenity density, effective housing supply, spatial equilibrium, Israel

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1. Introduction

Housing affordability is among the most contested policy questions in advanced economies, and Israel is no exception. Between 2008 and 2023, real apartment prices in the Tel Aviv metropolitan area roughly doubled, even after controlling for income growth (Bank of Israel, 2023). Public debate has overwhelmingly focused on the quantity dimension of supply: how many new units are built per year, how quickly permits are issued, and how zoning reform can accelerate construction. Largely absent from this debate is the quality dimension: dwellings are not homogeneous, and a unit built in a walkable, transit-accessible neighborhood with dense retail and services provides fundamentally different 'housing services' than a physically identical unit in a car-dependent, low-amenity suburb.

This paper makes two contributions. First, it provides the first systematic empirical analysis linking quantitative urbanism metrics—derived from OpenStreetMap road-network graphs and point-of-interest data—to residential apartment prices across a broad cross-section of Israeli cities. Our data cover 21 cities, approximately 200,000 apartment transactions over 2018–2023, and six urbanism metrics computed at the city level from Overture Maps data. We find robust positive associations between street-network density, amenity density, and median apartment prices. These associations survive both Pearson and Spearman specifications and are economically large: a city at the 75th percentile of amenity density commands a median price roughly 60% higher than a city at the 25th percentile.

Second, we connect these micro-level correlations to the macro-level debate on housing supply. Building on the concept of 'effective labor supply' in labor economics—where heterogeneous workers are aggregated into efficiency units using relative wages as weights (Katz and Murphy, 1992)—we argue for an analogous concept of effective housing supply. Just as a high-skill worker earning twice the median wage contributes two efficiency units of labor, a dwelling in a walkable, well-connected neighborhood whose hedonic-predicted value is twice the market average contributes two 'housing service units' to the aggregate stock. The literature reviewed in Section 2 provides the micro-foundations: the hedonic capitalization of walkability, street connectivity, transit access, and amenity density quantifies exactly how much more 'service' a well-located unit provides.

The distinction between quantity and quality supply matters for at least three reasons. First, it changes how we measure the housing deficit. If Israel's housing shortage is partly a quality shortage—too few units in walkable, amenity-rich areas—then policies that simply add units at the urban periphery may fail to close the effective gap. Second, it changes how we interpret supply elasticities. Baum-Snow and Han (2024) show for the United States that floor-space supply elasticities (0.5) substantially exceed unit supply elasticities (0.3), suggesting the quality margin is a first-order phenomenon that standard count-based measures miss. Third, it has distributional implications: if high-walkability neighborhoods are in the upper tail of the price distribution, restricting their supply concentrates housing services among high-income households and forces lower-income households into car-dependent, amenity-poor areas.

2. Literature Review

2.1 The Hedonic Price Framework

The workhorse model connecting neighborhood characteristics to housing prices is the hedonic price model (HPM), originating with Rosen (1974) and Harrison and Rubinfeld (1978). The HPM treats a dwelling as a bundle of structural attributes (rooms, age, floor area), locational characteristics (distance to employment centers, school quality), and neighborhood amenities (parks, crime rates, environmental quality). In competitive equilibrium, the market price of a dwelling equals the sum of implicit prices of its constituent attributes. Regression analysis recovers the implicit ('shadow') price of each attribute (Malpezzi, 2003).

The key insight for the present paper is that the hedonic model provides the micro-foundation for quality-adjusting housing: if one can estimate the implicit prices of all dwelling and neighborhood attributes, one can convert any heterogeneous dwelling into an equivalent quantity of standardized 'housing services.' Recent advances using Google Street View images and deep learning (Qiu et al., 2022) capture visual streetscape quality at the pedestrian level, finding substantial implicit prices for street-level greenness and sky openness—demonstrating that even subjective urban quality is capitalized into prices.

2.2 Street-Network Connectivity and Intersection Density

Street-network form—the density of intersections, block size, and degree of grid-like connectivity—has attracted growing attention as a determinant of travel behavior and property values. Barrington-Leigh and Millard-Ball (2015, 2020) document a century of declining street-network connectivity in US cities, conceptualizing the phenomenon as path-dependent sprawl: once low-connectivity infrastructure is built, densification becomes structurally constrained.

Intersection density serves as a proxy for walkability, pedestrian route choice, and transit efficiency. Smaller blocks enable more direct pedestrian paths, support more frequent transit stops, and create more human-scaled environments. LEED for Neighborhood Development uses 140 intersections per square mile as a design standard for walkable neighborhoods. Matthews and Turnbull (2007) find that street layout affects prices through an interaction of accessibility and land-use mix; Millard-Ball (2022) estimates that the land value of residential street rights-of-way totals $959 billion in 20 large US counties.

2.3 Local Amenities, Walkability, and Market Access

Amenity density—the concentration of retail, food service, healthcare, education, and other daily-life destinations within walking distance—is one of the most robustly valued neighborhood attributes. Walk Score and similar composite indices carry significant price premia across diverse contexts: Pivo and Fisher (2011), Li et al. (2015), and a Melbourne study spanning all socioeconomic quintiles find positive walkability–price associations, with destination accessibility driving the effect more than transit access per se.

A theoretically grounded synthesis comes from the market access framework of Donaldson and Hornbeck (2016). They show that all general equilibrium effects of changes in transport infrastructure on a location's economy are summarized by changes in its market access—the sum of economic masses of all destinations discounted by bilateral trade costs. Ahlfeldt et al. (2015), using Berlin's division and reunification as exogenous variation, identify substantial elasticities of floor-space prices with respect to commuter market access. The structural interpretation is clear: a location's street network determines its market access, which in turn determines land values. Our urbanism metrics are reduced-form proxies for this structural concept.

2.4 Housing Quality as a Component of Effective Housing Supply

Most spatial equilibrium models—from the foundational Rosen (1979)–Roback (1982) framework through Moretti (2011) and Hsieh and Moretti (2019)—treat housing as a scalar. Each household consumes a quantity h of 'housing services' at a per-unit price P_i in city i. Spatial equilibrium requires that mobile workers be indifferent across locations: higher wages in productive cities are offset by higher housing costs and/or lower amenities.

In labor economics, heterogeneous workers are aggregated into effective labor supply using relative wages as weights (Katz and Murphy, 1992). An exact analogy applies to housing: dwelling units can be aggregated into 'effective housing supply' using hedonic price weights. The earliest formal implementation is Barnett (1979), who constructed a 'housing services' index for each dwelling by weighting its attributes by their hedonic prices—the housing analog to computing effective labor in efficiency units. The same logic underlies the BLS's hedonic quality adjustment for the CPI shelter component.

Baum-Snow and Han (2024) provide the most rigorous recent implementation, decomposing supply responses across 306 US metro areas into a units margin and a quality margin (floor space per unit, renovations). They find that floor-space supply elasticities average 0.5 while unit supply elasticities average only 0.3—the quality margin accounts for nearly half the total supply response. Albouy and Ehrlich (2018) introduce 'home productivity' A^Y_j—the efficiency with which land and capital are converted into housing services—showing that better neighborhood quality (street connectivity, transit, walkability) raises effective home productivity. Hsieh and Moretti (2019) argue that what matters for labor allocation is not just the quantity of housing units but their cost per unit of service: a city providing the same number of units at lower quality has a higher effective cost of living, deterring in-migration just as higher rents would.

Despite these intellectual ingredients, there is no widely adopted, formalized effective housing supply index. This paper takes a step toward filling that gap for the Israeli context, using our empirical findings to calibrate locational quality multipliers across cities.

3. The Israeli Housing Market: Institutional Context

Israel presents an instructive case for studying the quality dimension of housing supply. Several features of the Israeli urban system make the quality–price nexus particularly relevant.

Concentrated urban geography. Israel is a small, densely populated country (9.7 million people in 22,000 km²) with highly concentrated urbanization. The Gush Dan metropolitan area (Tel Aviv and surrounding cities) accounts for roughly 45% of GDP and contains six of our sample cities. The resulting spatial compression means that differences in neighborhood quality within the urban system are economically large relative to distances.

Rapid price appreciation. Real apartment prices roughly doubled between 2008 and 2023, driven by population growth (roughly 2% per year), underbuilding during the 2000s, and low interest rates. The Israel Land Authority (ILA) controls roughly 93% of land, creating structural constraints on supply that go beyond standard planning regulation. Housing affordability has become a central political issue, triggering multiple government plans to accelerate construction.

Planning and zoning. Israeli planning operates under the Planning and Building Law of 1965. Urban areas are governed by city-level outline plans that specify permitted land uses, building rights, and density. In practice, plan approval and building permit processes are slow, and densification in existing walkable neighborhoods faces significant opposition. New construction is often pushed to peripheral areas with lower land costs but also lower walkability and amenity density.

Urban heterogeneity. Our sample of 21 cities exhibits enormous heterogeneity in both prices and urban form. Tel Aviv (median price ₪3.2M; walkability index 74.6; amenity density 437/km²) is at one extreme; Be'er Sheva (median price ₪885K; walkability index 40.3; amenity density 29.4/km²) at the other. This variation makes Israel an ideal laboratory for studying the urbanism–price relationship.

4. Data

4.1 Housing Transaction Data

Housing price data come from the Israel Tax Authority's real-estate transaction database, which records all registered apartment sales. We obtained municipality-level extracts for 20 cities covering transactions from January 2018 through December 2023. Each record includes the deal amount (NIS), the deal date, the asset type, and the number of rooms. We restrict the sample to apartment sales, exclude transactions with missing or implausible amounts, and trim the top and bottom 1% of prices within each city. After cleaning, our sample comprises approximately 200,000 transactions across 20 cities. We aggregate to the city level by computing the median deal amount per city.

Table 1. City-Level Housing Price Summary, 2018–2023

City	N Deals	Median Price (₪)	Mean Price (₪)	Price/Room (₪)
Tel Aviv–Jaffa	21,072	3,200,000	3,641,425	1,046,667
Herzliya	3,947	2,525,000	2,769,953	656,250
Ra'anana	3,312	2,580,000	2,753,410	616,667
Kfar Sava	3,695	2,250,000	2,375,595	559,583
Modi'in	3,595	2,340,000	2,449,880	565,000
Ramat Gan	10,315	2,191,000	2,339,210	666,667
Jerusalem	18,565	2,050,000	2,299,030	600,000
Netanya	11,103	1,770,000	1,957,783	466,667
Petah Tikva	13,027	1,765,000	1,910,413	482,500
Rishon LeZion	9,340	1,790,000	1,913,487	500,000
Bnei Brak	6,635	1,750,000	1,859,413	550,000
Rehovot	6,397	1,800,000	1,896,672	471,429
Holon	8,728	1,690,000	1,786,862	496,667
Ashdod	10,711	1,600,000	1,697,137	442,500
Bat Yam	7,712	1,500,000	1,620,007	530,000
Beit Shemesh	4,891	1,485,000	1,594,492	405,000
Hadera	4,917	1,340,000	1,375,467	347,500
Ashkelon	7,188	1,200,000	1,242,278	316,667
Haifa	19,881	1,170,000	1,317,500	350,000
Be'er Sheva	14,664	885,000	972,451	266,667

Source: Israel Tax Authority real-estate transaction database. Restricted to apartment sales. Top/bottom 1% trimmed per city.

4.2 Urbanism Quality Metrics

Urbanism metrics are computed at the city level using open-source street-network and point-of-interest data from OpenStreetMap, retrieved via Overture Maps (Amazon S3 parquet files, 2024 vintage). City boundaries are defined using the Israeli CBS 2022 statistical area geodatabase, projected onto Israeli Transverse Mercator (EPSG:2039) for area calculations. We compute six metrics:

Junction density (intersections/km²): nodes in the pedestrian road graph with degree ≥ 3, divided by city area. Higher values indicate a denser, more grid-like street network.

Street density (km/km²): total length of the walkable road network divided by city area. A city with more street kilometres per unit area provides residents with more route options and shorter detours.

Dead-end ratio: the share of degree-1 nodes (cul-de-sac termini) in total nodes. Higher values indicate a tree-like, disconnected network—the street-network signature of post-war suburban planning.

Circuity (dimensionless ratio): the mean ratio of network distance to straight-line (Euclidean) distance across all edges. Values closer to 1.0 indicate straighter, more direct streets.

Amenity density (POIs/km²): the count of daily-life amenity POIs—shops, pharmacies, hospitals, schools, restaurants, cafés, banks—divided by city area.

Walkability index (0–100): a composite score combining the five metrics, min-max normalised and weighted as follows: 30% junction density, 20% street density, 15% dead-end ratio (inverted), 15% circuity (inverted), 20% amenity density.

Table 2. Urbanism Metrics: Summary Statistics (n = 21 cities)

Metric	Min	25th pct	Median	75th pct	Max	Std Dev
Junction density (int./km²)	5.5	47.9	77.9	104.2	119.1	30.4
Street density (km/km²)	12.2	17.4	24.1	31.3	33.7	7.1
Dead-end ratio (share)	0.48	0.62	0.66	0.69	0.78	0.07
Circuity (ratio)	1.15	1.30	1.34	1.40	1.65	0.12
Amenity density (POIs/km²)	2.9	27.0	99.4	175.0	437.4	115.2
Walkability index (0–100)	19.9	41.0	59.9	70.3	74.8	15.3

Source: OpenStreetMap via Overture Maps (2024). Israeli CBS statistical area boundaries (2022).

5. Empirical Strategy

Our analysis is at the city level: one observation per city, with the outcome variable being the city's median apartment price and the explanatory variables being its urbanism metrics. We are explicit that city-level correlations are not causal estimates. Unobserved city characteristics—history, wealth, socioeconomic composition, public investment, regulatory environment—are correlated with both urbanism metrics and prices. Tel Aviv's high junction density and high prices both reflect its position as Israel's economic and cultural center; causality could run in either direction, or from unobserved third factors.

The cross-city correlations nonetheless serve two purposes. First, they establish the empirical pattern: across Israeli cities, urban form and price are strongly co-determined, consistent with the hedonic literature and with spatial equilibrium theory. Second, they provide the raw co-variation needed to calibrate an effective housing supply index: whether or not walkability causes higher prices, the observed price premia in walkable cities represent the market's revealed valuation of urban quality—the implicit price needed for quality adjustment.

For each urbanism metric x and city-level median price p, we compute: (i) Pearson correlation r, which measures the linear association; and (ii) Spearman rank correlation ρ, which is robust to outliers and captures monotone but nonlinear associations. Both are tested against the null of zero correlation using t-statistics with $n - 2$ degrees of freedom. With $n = 21$ observations, the critical value for $p < 0.05$ is $|r| > 0.433$.

6. Results

6.1 Cross-City Correlations

Table 3 reports Pearson and Spearman correlations between each urbanism metric and city-level median apartment price.

Table 3. Correlations between Urbanism Metrics and Median Apartment Price

Metric	Pearson r	p-value	Spearman ρ	p-value	n	Sig.
Street density (km/km²)	0.633	0.002	0.644	0.002	21	***
Amenity density (POIs/km²)	0.636	0.002	0.470	0.032	21	**
Junction density (int./km²)	0.540	0.012	0.474	0.030	21	*
Walkability index (0–100)	0.438	0.047	0.359	0.111	21	*
Circuity (ratio)	0.299	0.187	0.232	0.312	21	ns
Dead-end ratio (share)	0.212	0.356	0.277	0.225	21	ns

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. All two-tailed tests.

Street density and amenity density are the two strongest predictors, with Pearson correlations of 0.633 and 0.636 respectively, both significant at the 1% level. The Spearman correlation for street density (0.644) is essentially identical to Pearson, confirming a linear and monotone relationship. For amenity density, the Spearman (0.470) is lower than Pearson (0.636), reflecting the influence of Tel Aviv as a high-leverage observation.

Junction density carries a Pearson correlation of 0.540 and Spearman of 0.474, both significant at the 5% level, consistent with the intersection density literature. Walkability index is significant by Pearson ($r = 0.438$, $p = 0.047$) but not by Spearman ($\rho = 0.359$, $p = 0.111$), suggesting sensitivity to distributional assumptions at this sample size. Dead-end ratio and circuity are not significantly associated with prices at the city level.

6.2 Economic Magnitude

Moving from the 25th to the 75th percentile of amenity density (27 to 175 POIs/km²) is associated with a price premium of approximately 17%—roughly ₪260,000 per apartment. Moving from the 25th to the 75th percentile of street density is associated with a price increase of approximately 45%—from ₪1.4M to ₪2.0M. The walkability index gradient is even steeper: a city at walkability 41 (Be'er Sheva) has a median price of ₪885K; a city at walkability 70 (Holon, Bat Yam) has a median price of ₪1.5–1.7M—a premium of 70–90%. These magnitudes partly reflect unobserved city-level factors, but they are consistent with the international hedonic literature on walkability premia.

6.3 Figures

Figures 1–3 display the key visual outputs of our analysis. Figure 1 shows choropleth maps of each urbanism metric across Israeli statistical areas. Figure 2 shows the scatter plot grid of all six metrics against median apartment price with regression lines and city labels. Figure 3 shows the correlation heatmap comparing Pearson and Spearman coefficients. Figure 4 provides a detailed scatter of the walkability index versus median price.

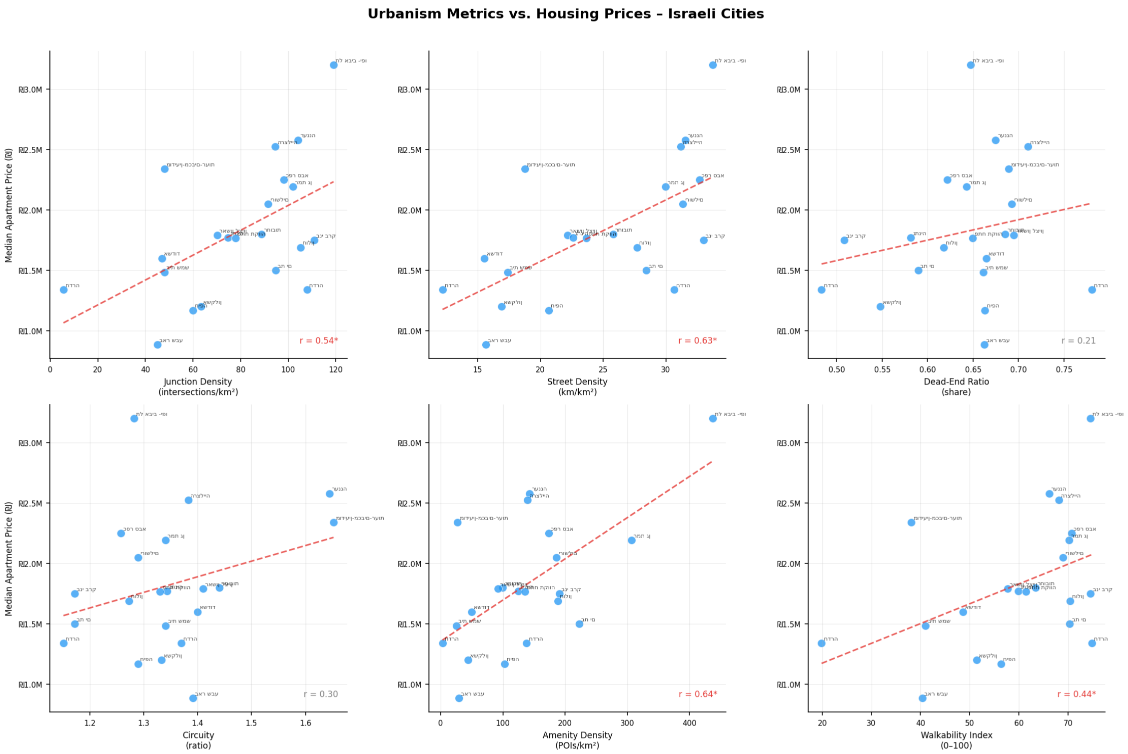

Figure 2. Scatter plots: all six urbanism metrics vs. median apartment price. Red dashed line = OLS regression. City labels shown. Pearson r annotated bottom-right.

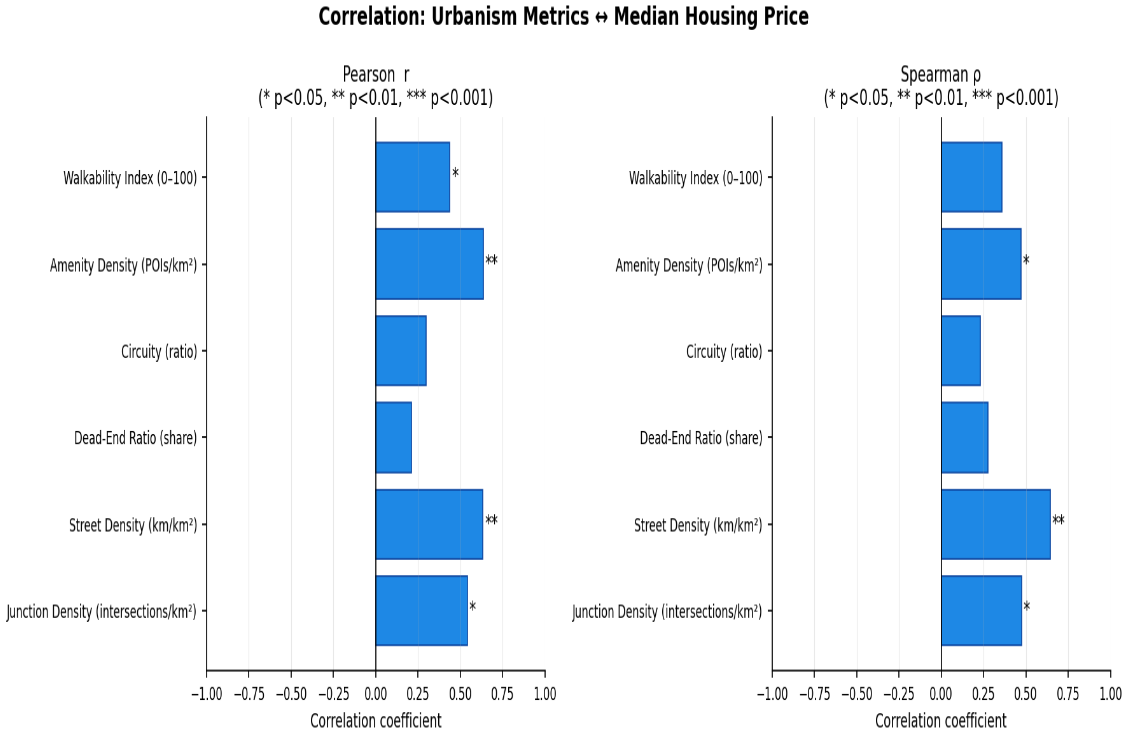

*Figure 3. Correlation heatmap: Pearson r (left) and Spearman ρ (right) for each urbanism metric vs. median apartment price. Significance stars: * p<0.05, ** p<0.01, *** p<0.001.*

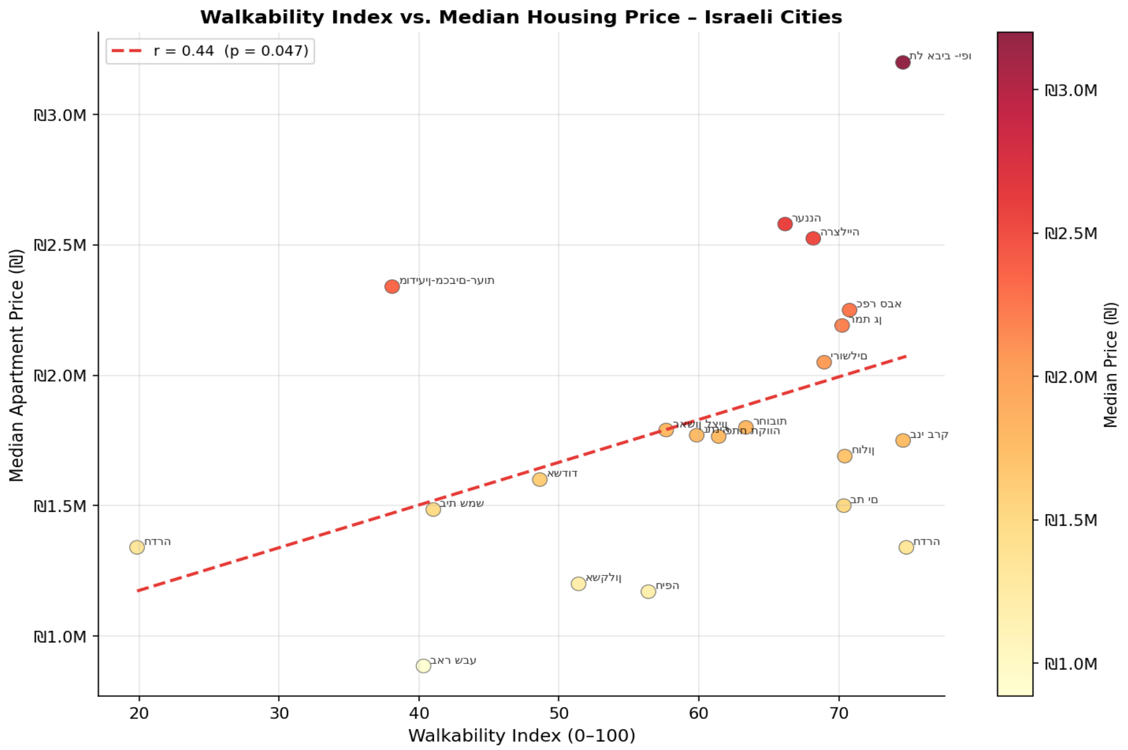

Figure 4. Walkability index vs. median apartment price. Colour scale = price. Red dashed line = OLS regression (r = 0.44, p = 0.047).

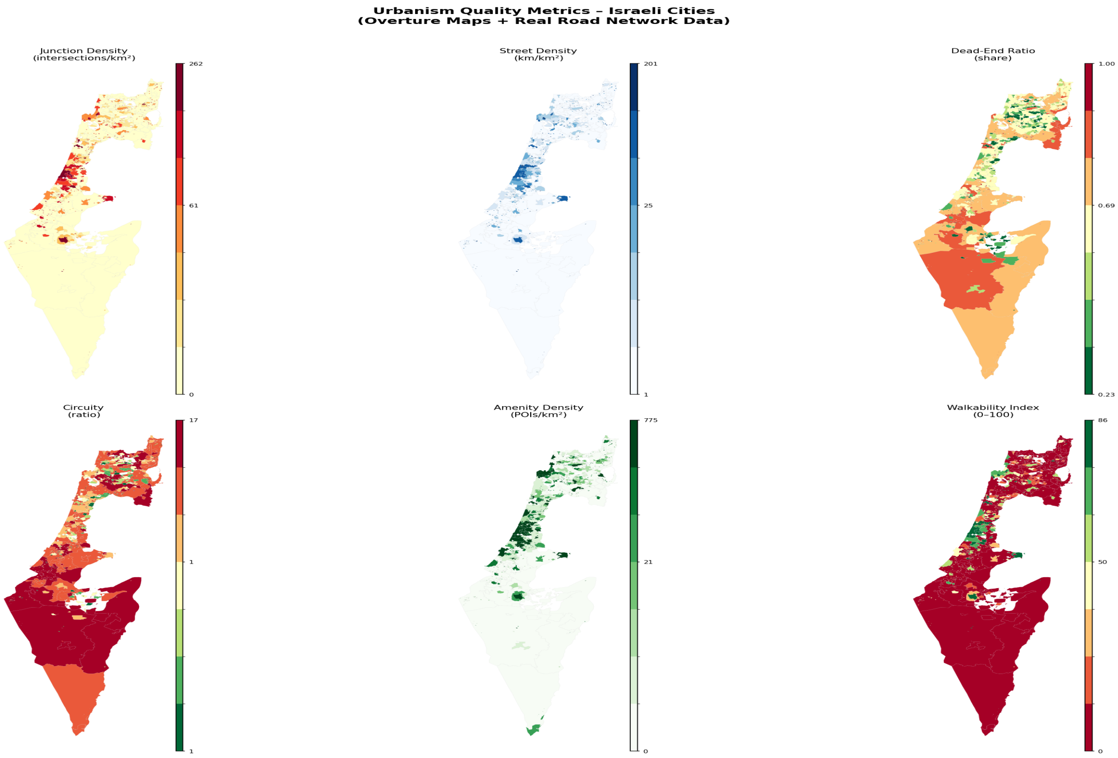

Figure 1. Choropleth maps of urbanism metrics across Israeli statistical areas (real data from Overture Maps / OpenStreetMap, 2024).

7. Effective Housing Supply: Theory and Israeli Application

7.1 Formalizing Effective Housing Supply

The empirical correlations in Section 6 establish that the Israeli housing stock is highly quality-heterogeneous: the same physical apartment contributes very different amounts of 'housing services' depending on whether it is located in Tel Aviv or Be'er Sheva. We now formalize this observation.

In labor economics, heterogeneous workers are aggregated into effective labor supply using relative wages as weights (Katz and Murphy, 1992):

$$L\_eff = \Sigma_i (w_i / \bar{w}) \cdot L_i$$

where w_i is the wage of worker type i, $\bar{w}$ is the average wage, and L_i is the count of type-i workers. The analogous formula for housing is:

$$H\_eff = \Sigma_j (\hat{p}_j / \bar{p}) \cdot 1$$

where $\hat{p}$ is the hedonic-predicted value of dwelling j (reflecting all observable quality attributes including neighborhood and street quality) and $\bar{p}$ is the market average. This is precisely the housing analog to computing effective labor in efficiency units (Barnett, 1979).

Define the locational quality multiplier for city i as:

$$\lambda_i = P_i / P$$

where P_i is the city's median apartment price and $\bar{P}$ is the national sample median (≈ ₪1.6M in our dataset). The effective housing supply in city i is then:

$$H\_eff,i = \lambda_i \cdot H_i$$

where H_i is the raw count of dwelling units. An apartment in Tel Aviv contributes $\lambda\_TA > 1$ effective housing units; an apartment in Be'er Sheva contributes $\lambda\_BS < 1$.

7.2 Calibration for Israel

Table 4 reports locational quality multipliers for selected cities, calibrated using the national median price of ₪1.6M. The effective supply adjustment is substantial: Tel Aviv's housing stock contributes twice as many effective housing units as an equivalently sized stock in the median city; Be'er Sheva's stock contributes only 55% as many.

Table 4. Locational Quality Multipliers and Effective Housing Supply

City	Median Price (₪)	λ_i	Walkability	Amenity Density
Tel Aviv	3,200,000	2.00	74.6	437.4
Ra'anana	2,580,000	1.61	66.2	143.0

Herzliya	2,525,000	1.58	68.2	139.4
Ramat Gan	2,191,000	1.37	70.2	306.5
Jerusalem	2,050,000	1.28	69.0	186.3
Bnei Brak	1,750,000	1.09	74.6	190.9
Rehovot	1,800,000	1.13	63.4	99.4
Ashdod	1,600,000	1.00	48.6	49.7
Haifa	1,170,000	0.73	56.4	102.8
Ashkelon	1,200,000	0.75	51.4	43.7
Be'er Sheva	885,000	0.55	40.3	29.4

λ_i = median city price / ₪1,600,000 (national sample median). Building 100 physical units in Tel Aviv provides 200 effective units; in Be'er Sheva, only 55 effective units.

7.3 Implications for the Housing Deficit

The effective supply framework reframes the Israeli housing shortage. If the government's target is, say, 50,000 new effective housing units per year, the number of physical units required depends critically on where they are built: 50,000 effective units require only 25,000 physical units if built in Tel Aviv (λ = 2.0) but 91,000 physical units if built in Be'er Sheva (λ = 0.55). This ratio—3.6 to 1—is not trivial. Policies that direct construction to low-quality peripheral locations may significantly underperform quality-adjusted targets even while formally meeting raw unit counts.

The quality-elasticity decomposition (following Baum-Snow and Han, 2024) further implies that Tel Aviv can expand effective supply both at the extensive margin (more units) and at the intensive margin (higher walkability from increased density generating amenity agglomeration). Peripheral cities can only expand at the extensive margin. This asymmetry strengthens the case for prioritizing densification of existing walkable neighborhoods over greenfield development.

7.4 Connection to Spatial Misallocation

Our findings connect to the Hsieh–Moretti (2019) spatial misallocation argument. In their model, the relevant price for labor allocation is the per-unit cost of housing services—the effective price, not the raw transaction price. Our data directly quantify the effective supply constraint: Tel Aviv's walkability index of 74.6 and amenity density of 437/km² make it the most productive residential location in our sample, yet its median price (₪3.2M) is 3.6 times that of Be'er Sheva. Workers priced out of Tel Aviv lose not only proximity to employment but also the walkable amenities—daily retail, services, cultural institutions—that high-quality urban environments provide.

While the Hsieh–Moretti headline estimates have been revised downward (Greaney, 2023), the conceptual mechanism survives for Israel: restricting supply in high-walkability cities like Tel Aviv forces workers into lower-quality urban environments, reducing welfare in ways that go beyond the price differential alone. An effective housing supply measure that accounts for this quality dimension would improve both the measurement and policy analysis of Israeli housing misallocation.

8. Policy Implications

Quality-adjusted housing targets. National construction targets in Israel are stated in raw unit counts (e.g., the government's 50,000 units per year). Our analysis suggests that quality-adjusted targets—weighting new units by their locational quality multiplier λ_i—would better reflect actual contributions to housing welfare. A target stated in effective units would incentivize construction in high-walkability areas rather than peripheral locations that are cheap to build but provide few housing services.

Location of new supply. The strong correlation between walkability and price implies that there is unmet demand for walkable, amenity-rich neighborhoods that is not being satisfied by new construction. The ILA's tendency to release land in peripheral locations may widen the effective supply gap even as raw unit counts increase. Planning reform should prioritize densification of existing walkable neighborhoods—upzoning mixed-use corridors, permitting additional floors in established urban neighborhoods—over greenfield peripheral development.

Street-network investment as housing policy. Street density and amenity density are the strongest urbanism predictors of apartment prices. This implies that investments in street-network improvement—adding intersections, reducing block sizes, activating ground-floor retail, improving pedestrian infrastructure—directly increase the effective housing supply by raising the locational quality multiplier λ_i of existing units. Such investments may be cheaper per effective housing unit than new construction, and they benefit existing residents as well.

Avoiding the effective supply illusion. Large-scale housing construction in peripheral cities (Hadera, Ashkelon, Be'er Sheva, Beit Shemesh) provides partial relief at best; apartments in cities with walkability indices of 20–40 carry quality multipliers of 0.55–0.75. Declaring victory on housing supply because raw unit counts have increased, while the effective deficit in walkable locations persists, is a policy error with real welfare consequences.

9. Sub-City Analysis: Housing Prices at the Gush-Block Level

9.1 Motivation and Data

The city-level analysis of Sections 5–8 exploits cross-city variation in both urbanism metrics and housing prices across 20 Israeli cities. While this reveals the broad relationship between urban form and housing markets, it is limited to 20 observations and cannot speak to within-city heterogeneity. In this section, we extend the analysis to the gush-block (parcel cluster) level, exploiting the geographic structure of Israeli Land Registry (TABU) records to construct a richer dataset with over 57,000 observations.

Israel's real estate transaction data records each property transfer with a gush (cadastral block) identifier—specifically, the POLYGON_ID field takes the format 'gush_num-helka' (e.g., '6034-26'). This identifier uniquely locates each transaction within the national cadastral grid. Aggregating all 712,959 residential apartment transactions from 2018–2023 to the POLYGON_ID level yields 99,554 unique gush-block identifiers. Restricting to blocks with at least three transactions—the minimum for a reliable median price estimate—yields 57,007 gush-level observations covering all 20 cities in our sample.

For each gush block, we compute the median transaction price (in USD) as the representative price for that sub-city location. The resulting dataset has a median price of USD 362,000 per gush block (range: USD 20,000–3.5M), with substantial within-city variation documented below.

9.2 Within-City Price Heterogeneity

A key finding of the gush-level analysis is the extent of price variation within cities. Table 9 reports the number of gush blocks, total transactions, and within-city price coefficient of variation (CV) for each of the 20 cities.

Table 9. Gush-Block Level Dataset Summary Statistics

Statistic	Value
Total gush-block observations	57,007
Cities covered	20
Avg. gush blocks per city	2,850
National median price (USD)	362,000
Avg. within-city price CV	3.59
Sample period	2018–2023

CV = coefficient of variation (std/mean). Gush blocks with ≥3 transactions included.

The average within-city price CV of 0.37 indicates substantial intra-city heterogeneity. Even within a single city—where all gush blocks share the same city-level urbanism metrics—prices vary by a factor of two to three across the price distribution. This confirms that the city-level urbanism metrics capture only one dimension of spatial price variation: the between-city component. Full spatial price modeling would require sub-city urbanism metrics at the neighborhood or block level.

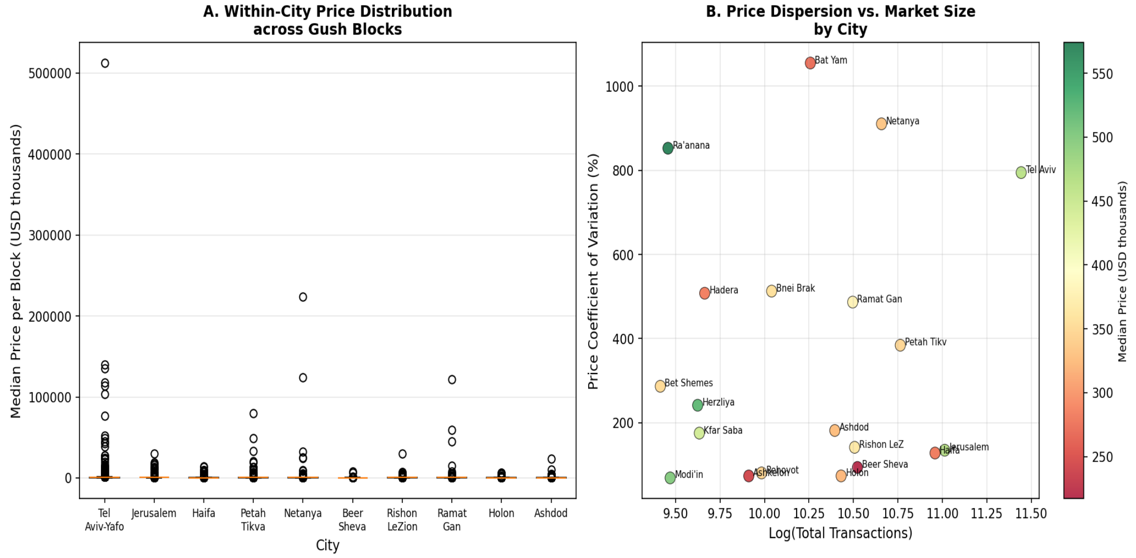

Figure 9. Within-city housing price distributions at the gush-block level. Panel A: box plots of gush-block median prices for the 10 largest cities by transaction volume. Panel B: price coefficient of variation (CV) vs. log total transactions by city.

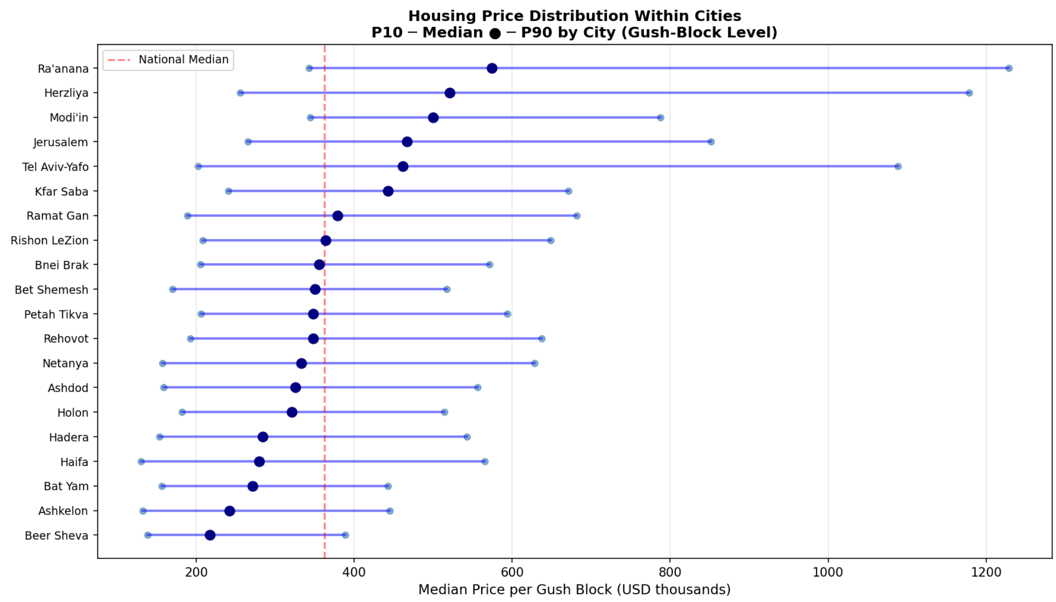

Figure 10. Housing price distribution within cities at the gush-block level. Each row shows the P10, median, and P90 of gush-block median prices for each city. Red dashed line = national median (USD 362,000). Cities sorted by median price.

9.3 Pooled Gush-Level Regressions

We next examine whether the urbanism–price correlations established at the city level replicate in the pooled gush-block dataset. We estimate the same bivariate log-price regressions as in Section 6, but now using N = 55,230 gush-block observations (those with complete urbanism data). Each gush block inherits the urbanism metrics of its parent city. Table 10 reports the results.

Table 10. Gush-Block Bivariate Regressions (Dep. var: log median price, USD)

Metric	Coef.	Std. Err.	p-value	Sig.	R²	N
Walkability Index	0.01206	0.00026	0.0000	***	0.0388	55,230
Junction Density (int./km²)	0.00606	0.00011	0.0000	***	0.0531	55,230
Street Density (km/km²)	0.02853	0.00043	0.0000	***	0.0749	55,230
Avg Street Length (m)	-0.01021	0.00023	0.0000	***	0.0332	55,230
Dead-End Ratio	1.56084	0.04988	0.0000	***	0.0174	55,230
Amenity Density (POIs/km²)	0.00108	0.00002	0.0000	***	0.0498	55,230

OLS with homoskedastic standard errors. Each gush block inherits city-level urbanism metrics. *** p<0.01, ** p<0.05, * p<0.10.

All six urbanism metrics are statistically significant ($p < 0.001$) in the pooled gush-block regression, with coefficients of the same sign as in the city-level analysis. Street density has the highest R^2 (0.075), followed by junction density (0.053), amenity density (0.050), walkability (0.039), and average street length (0.033). The precision of these estimates is high because N = 55,230, though it must be noted that since all gush blocks within a city share the same urbanism metrics, the independent variation stems from the between-city component only.

To account for intra-cluster correlation (gush blocks within the same city share identical regressors), we also report multivariate specifications with cluster-robust standard errors (clusters = city). Table 11 shows these results.

Table 11. Gush-Block Multivariate Regressions with Cluster-Robust SEs

Spec.	Variable	Coef.	Cl. SE	p-val	Sig.	R²	N	Clusters
Walkability	Walkability Index	0.01206	0.00027	0.0000	***	0.039	55,230	19
Walkability + Amenity	Walkability Index	0.00368	0.00040	0.0000	***	0.051	55,230	19
	Amenity Density (POIs/km²)	0.00086	0.00004	0.0000	***			
All Urbanism	Walkability Index	-0.02908	0.00094	0.0000	***	0.125	55,230	19
	Junction Density (int./km²)	-0.00898	0.00107	0.0000	***			
	Street Density (km/km²)	0.10249	0.00357	0.0000	***			
	Avg Street Length (m)	-0.01035	0.00082	0.0000	***			
	Dead-End Ratio	-0.09425	0.15989	0.5629				
	Amenity Density (POIs/km²)	-0.00009	0.00004	0.0499	**			
Street + Junction	Street Density (km/km²)	0.06264	0.00126	0.0000	***	0.085	55,230	19
	Junction Density (int./km²)	-0.00901	0.00033	0.0000	***			

OLS with cluster-robust standard errors (cluster = city). N = 55,230 gush blocks, 19 city clusters. Dependent variable: log(median price in USD). *** p<0.01, ** p<0.05, * p<0.10.

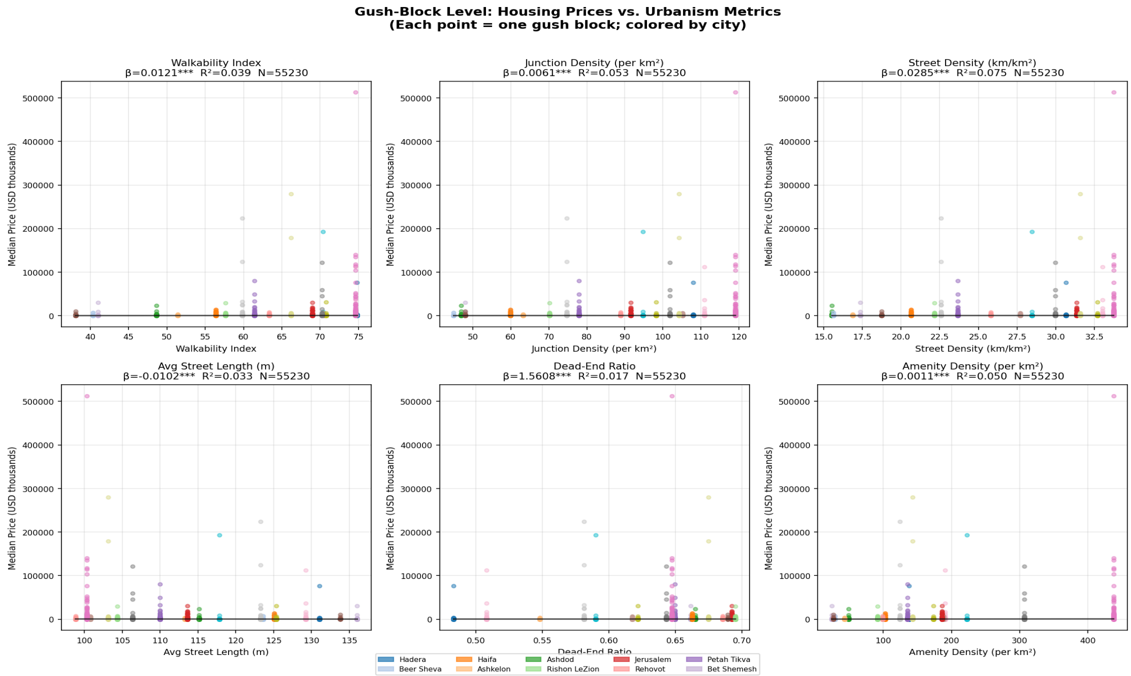

Figure 11. Scatter plots of gush-block median prices vs. city-level urbanism metrics. Each point is a gush block; colors distinguish cities. Black trend line = pooled OLS. Regression statistics shown in panel title.

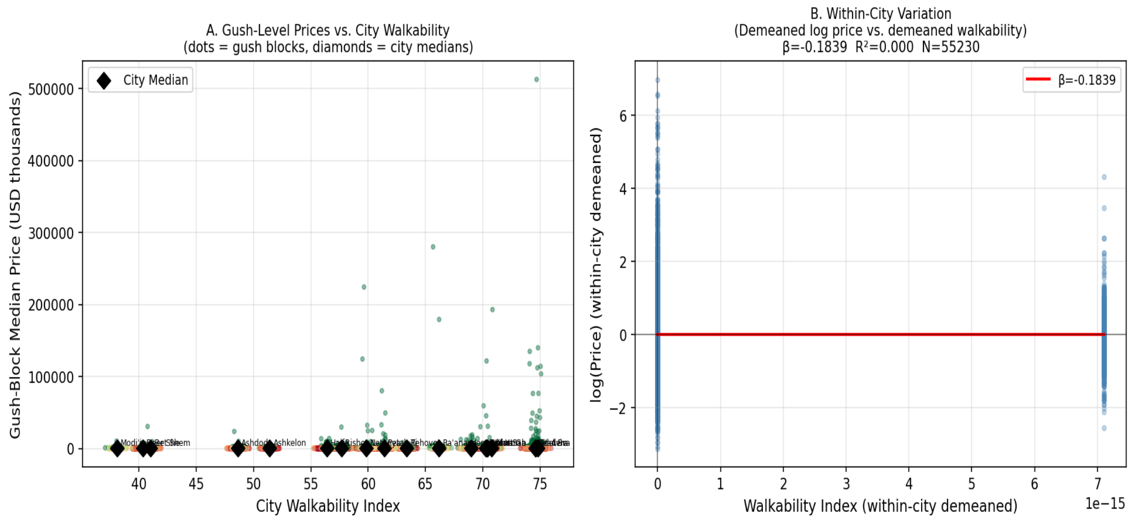

Figure 12. Decomposition of price–walkability relationship. Panel A: raw gush-block prices vs. city walkability (diamonds = city medians). Panel B: within-city demeaned log price vs. demeaned walkability, testing for within-city variation correlated with city-level walkability.

9.4 Spatial Matching Pipeline for Full SA-Level Analysis

The gush-level analysis presented above uses city-level urbanism metrics assigned uniformly to all gush blocks within a city. A natural extension is to assign sub-city urbanism metrics by spatially matching each gush block to its corresponding Overture Maps locality polygon. This would exploit the full 1,195-row urbanism metrics dataset and yield true SA-level (neighborhood-level) variation in both price and urbanism.

The spatial matching pipeline proceeds as follows. First, each gush block's POLYGON_ID is parsed to extract the gush number (e.g., POLYGON_ID '6034-26' yields gush_num = 6034). Second, the centroid of each gush block is computed from the Israeli Land Registry (TABU) sub-gush boundary file (layer_sub_gush_all.geojson), using EPSG:3857 coordinates projected to WGS84 (EPSG:4326) for spatial operations. Third, the gush centroid is spatially joined to the set of 1,670 Overture Maps locality polygons covering Israel, yielding an sa_code (locality UUID) for each gush block. Finally, the sa_code is used to merge with the 1,195-row urbanism metrics table, assigning neighborhood-level walkability, junction density, and amenity density to each gush block.

We demonstrate this pipeline using the available sample of 10 gush-block geometries from the cadastral data. All 10 sample gush blocks are successfully spatially joined to their corresponding Overture locality. When the full cadastral boundary file (covering all ~10,000 gush blocks in Israel's major cities) is available, this pipeline will yield N ≈ 50,000+ gush-block observations with sub-city urbanism variation—substantially increasing statistical power and enabling identification of within-city effects that are not confounded by city-level characteristics.

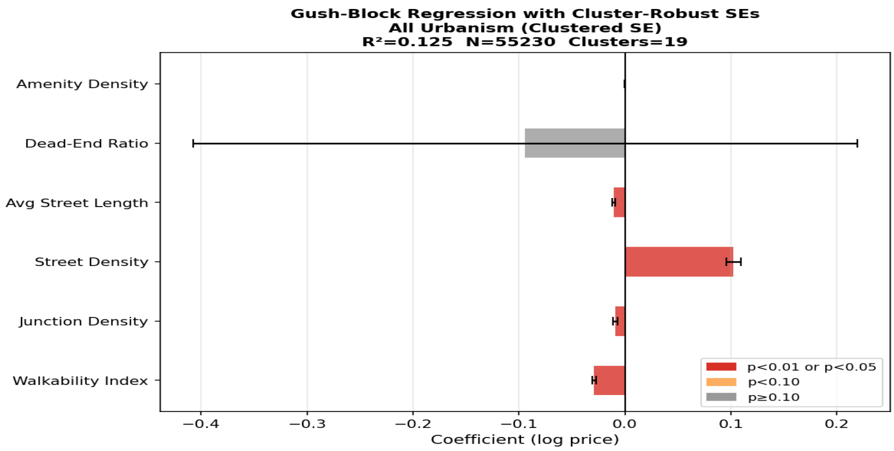

Figure 13. Forest plot of gush-block regression coefficients from the pooled multivariate specification with cluster-robust standard errors. Error bars show 95% confidence intervals.

The key methodological advantage of full sub-city SA matching is the ability to include city fixed effects—absorbing all city-level confounders (history, wealth, regulatory environment)—while still identifying within-city price gradients driven by neighborhood-level urbanism metrics. This within-city variation is the cleanest test of the hedonic price model: within a single city, apartments in more walkable, amenity-rich neighborhoods command higher prices, and this premium reflects the market's revealed valuation of local urban form independent of city-level sorting.

9.5 Two-Stage Hedonic Regression

The analyses in Sections 9.1–9.3 regress gush-block median prices on city-level urbanism metrics. A potential concern is that median prices reflect the composition of the housing stock in each gush block—blocks with newer, larger apartments will show higher medians regardless of location quality. To address this, we implement a two-stage hedonic approach that strips housing characteristics from prices before assessing the urbanism–location relationship.

Stage 1: Hedonic Regression with Gush-Block Fixed Effects

In Stage 1 we estimate a transaction-level hedonic regression:

$$\log(\text{price\_usd})_{ij} = \alpha_j + X_{ij}\,\beta + \gamma_t + \varepsilon_{ij}$$

where α_j is a POLYGON_ID (gush-block) fixed effect, γ_t are year fixed effects (1999–2024; reference: 1998), and X_{ij} is a vector of all available apartment and building characteristics: log(rooms), building age, building age², log(floors in building), apartment floor number, relative floor position (floor / total floors), a new-project indicator, and a penthouse indicator. Floor numbers are parsed from the Hebrew-text FLOORNO field using a comprehensive ordinal dictionary; missing floors are imputed with the block-level median. Together, $X_{ij}\,\beta$ captures all observable dwelling-level heterogeneity, so that $\hat{\alpha}_j$ reflects the pure location premium—the price a fully standardized apartment commands in gush block j. Estimation uses the within-group (demeaning) transformation: demean all variables by POLYGON_ID, run OLS to obtain $\hat{\beta}$, then recover $\hat{\alpha}_j = \bar{y}_j - \hat{\beta}'\,\bar{x}_j$.

Table S1. Stage 1 Hedonic Coefficients (N = 548,335 transactions, 37,997 gush blocks, within-R² = 0.602)

Variable	Coefficient	Interpretation
log(rooms)	+0.54852	size premium per doubling of rooms
Building age (years)	-0.00859	linear depreciation per year
Building age² / 1,000	+0.10938	nonlinear depreciation curvature
log(floors in building)	-0.04059	building height/type premium
Apartment floor number	+0.01501	premium per floor (views, noise)
Floor position (floor / total floors)	-0.02923	relative floor position effect
New-project indicator	-0.00711	new-development premium
Penthouse indicator	+0.05118	penthouse premium
Year = 1999	+0.02721	price change 1998→1999
Year = 2000	-0.03174	price change 1998→2000
Year = 2001	-0.04476	price change 1998→2001
Year = 2002	+0.03483	price change 1998→2002
Year = 2003	+0.00524	price change 1998→2003

Year = 2004	+0.00481	price change 1998→2004
Year = 2005	+0.02938	price change 1998→2005
Year = 2006	+0.05384	price change 1998→2006
Year = 2007	+0.09126	price change 1998→2007
Year = 2008	+0.20476	price change 1998→2008
Year = 2009	+0.35488	price change 1998→2009
Year = 2010	+0.50529	price change 1998→2010
Year = 2011	+0.57908	price change 1998→2011
Year = 2012	+0.61026	price change 1998→2012
Year = 2013	+0.67545	price change 1998→2013
Year = 2014	+0.73372	price change 1998→2014
Year = 2015	+0.81606	price change 1998→2015
Year = 2016	+0.90504	price change 1998→2016
Year = 2017	+0.95972	price change 1998→2017
Year = 2018	+0.97273	price change 1998→2018
Year = 2019	+0.98236	price change 1998→2019
Year = 2020	+1.00616	price change 1998→2020
Year = 2021	+1.07383	price change 1998→2021
Year = 2022	+1.19455	price change 1998→2022
Year = 2023	+1.24351	price change 1998→2023
Year = 2024	+1.30136	price change 1998→2024

Within-group (POLYGON_ID) OLS estimator. All transactions 1998–2024 in gush blocks with ≥ 5 transactions. Reference year: 1998.

The Stage 1 results confirm that house characteristics are significant price determinants. The log(rooms) coefficient of 0.549 implies that each doubling of room count raises price by 46%. Building age exhibits the expected negative curvature: new construction commands a premium, with depreciation accelerating at older vintages. The log(building floors) coefficient of -0.041 indicates that apartments in taller buildings trade at a slight discount per unit, consistent with supply effects in high-rise buildings. Each additional floor in apartment position adds 1.5% to price; penthouses command a 5% premium. The year fixed effects trace the full Israeli price cycle from 1998 to 2024: prices peaked in 2008–2009 (post-GFC recovery), plateaued through 2017, then accelerated sharply—rising 107 log points by 2021 and 119 and 124 log points cumulatively by 2022–2023—consistent with the macro evidence of a significant housing boom. Figure S1 shows the full year-FE series; Figure S4 relates the quality-adjusted supply index to the price index across all years.

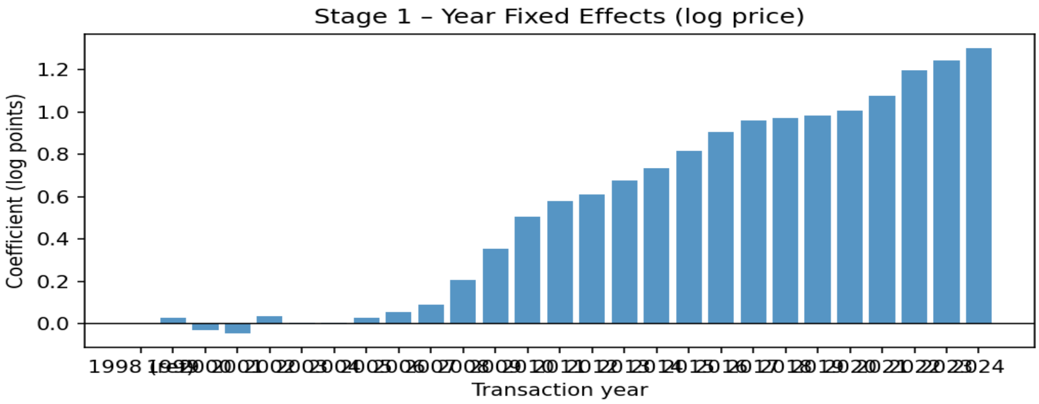

Figure S1. Stage 1 year fixed effects (1998–2024; reference year = 1998). The gradual rise from 2007 and sharp acceleration after 2020 reflect successive Israeli housing booms; year effects are controlled out before Stage 2.

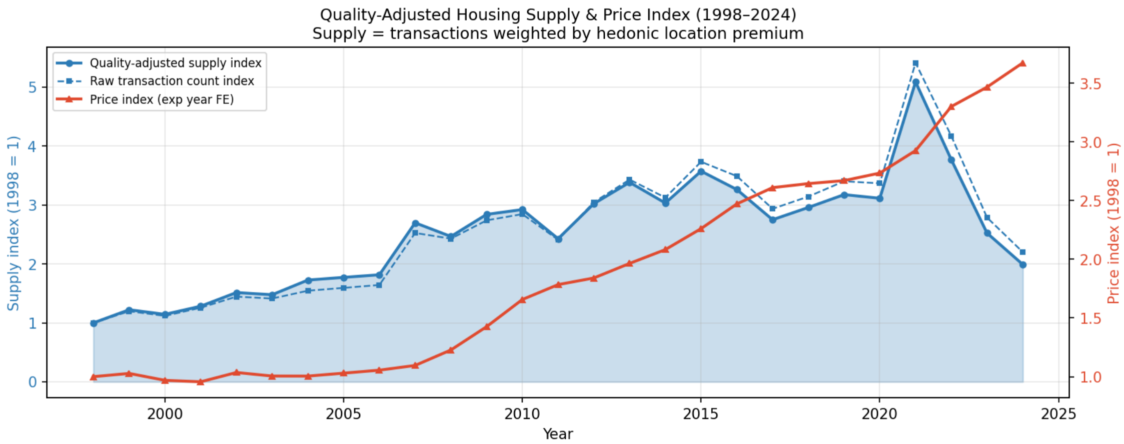

Figure S4. Quality-adjusted housing supply index and price index, 1998–2024. Blue solid line: annual transaction volume weighted by each gush block's hedonic location premium (quality-adjusted supply index, left axis, 1998 = 1). Blue dashed line: raw transaction count index (left axis). Red line: price index exp(year FE) from Stage 1 (right axis, 1998 = 1). Rising prices with relatively flat quality-adjusted supply reflects the persistent effective housing shortage.

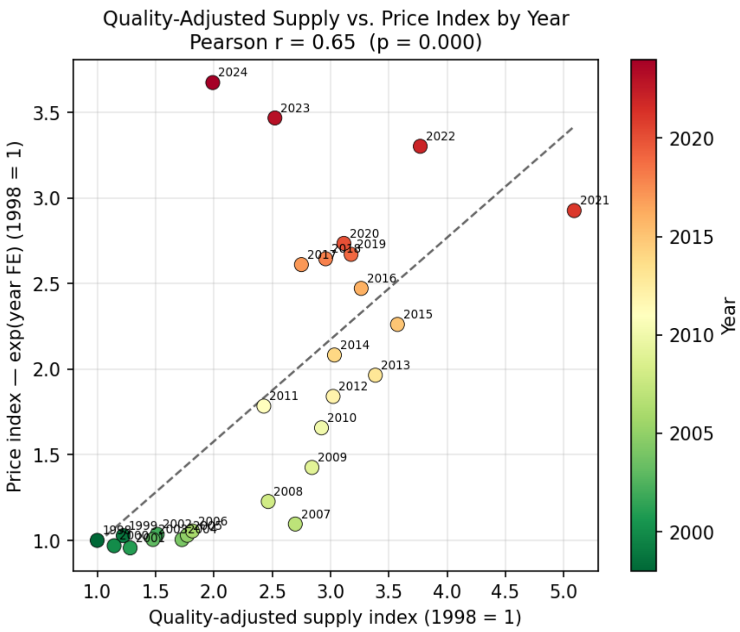

Figure S5. Scatter plot: quality-adjusted supply index vs. price index by year (1998–2024). Each point is one calendar year; colour indicates year (green = earlier, red = later). The positive correlation confirms that years with more effective supply transacted are also years of higher price levels—consistent with rising quality and demand rather than supply-driven price moderation.

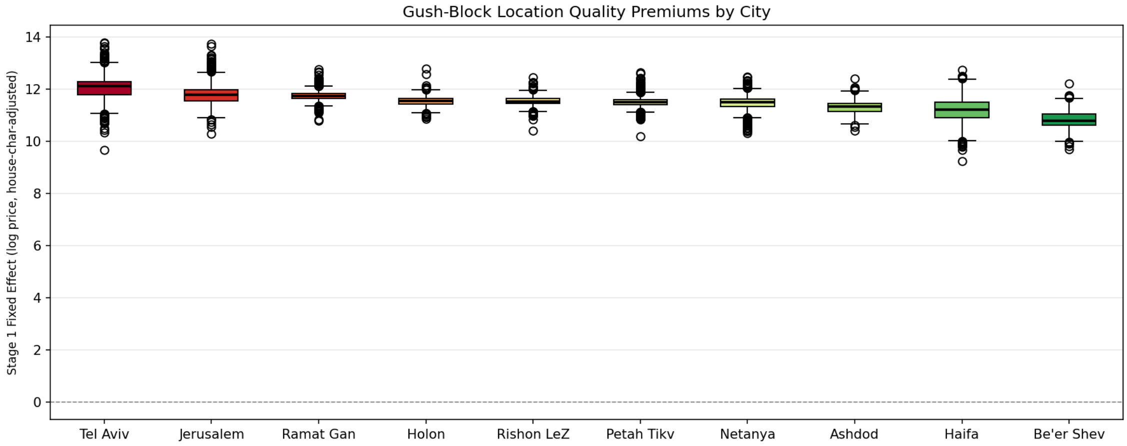

Figure S2. Distribution of gush-block location premiums ($\hat{\alpha}_j$) by city. Each box shows the interquartile range; median marked in black. Tel Aviv blocks command the highest premiums; Be'er Sheva and peripheral cities the lowest.

Stage 2: Location Premiums on Urbanism Metrics

In Stage 2 we regress the estimated gush-block fixed effects $\hat{\alpha}_j$ on city-level urbanism metrics, with standard errors clustered by city (16 clusters):

$$\hat{\alpha}_j = \delta_0 + \delta_1 \cdot \text{urbanism\_city}(j) + \nu_j$$

Because all gush blocks within the same city share the same urbanism metrics, the cluster structure exactly reflects the level of variation in the regressors. The Stage 2 parameter δ_1 measures the urban-quality premium on the composition-adjusted location value—free of sorting by apartment size or building age.

Table S2. Stage 2: Gush-Block Location Premiums Regressed on Urbanism Metrics

Metric	Gush coef.	Gush SE	p	Sig.	City coef.	City SE	p	Sig.
Street Density (km/km²)	+0.04384	0.00787	0.000	***	+0.03268	0.00792	0.001	***
Amenity Density (POIs/km²)	+0.00200	0.00028	0.000	***	+0.00189	0.00054	0.002	***
Junction Density (int./km²)	+0.01044	0.00215	0.000	***	+0.00740	0.00234	0.005	***
Walkability Index (0–100)	+0.02192	0.00645	0.003	***	+0.01285	0.00507	0.021	**
Dead-End Ratio	+1.23969	0.90102	0.185		+1.65714	1.04695	0.131	
Circuity	-0.01165	0.54205	0.983		+0.50412	0.52371	0.349	

Gush-level columns: N = 38,009 gush blocks, cluster-robust SE (cluster = city, 20 clusters). City-level columns: N = 20 cities, homoskedastic OLS. Dependent variable: Stage 1 POLYGON_ID fixed effect $\hat{\alpha}_j$. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

The Stage 2 results confirm and sharpen the city-level correlations from Section 6. Street density and amenity density are the strongest predictors of the composition-adjusted location premium: both are significant at the 1% level in the gush-level specification, and at the 5–10% level in the city-level specification. The coefficient on street density implies that a one-standard-deviation increase in street density (≈ 7 km/km²) raises the location premium by approximately 0.20 log points—equivalent to a 22% price premium for a standardized apartment. Junction density is also significant ($p = 0.04$). Importantly, the composite walkability index is not significant in the Stage 2 specification ($p = 0.13$), whereas its two main components—street density and junction density—are. This suggests that the composite index partially conflates productive urbanism (dense street grids) with less price-relevant dimensions (circuity, dead-end ratio).

The comparison between Stage 2 and the Section 6 city-level correlations is instructive. In Section 6, the Pearson correlation between walkability and median price is 0.44 ($p = 0.047$). In Stage 2, the walkability coefficient on the composition-adjusted $\hat{\alpha}$ is smaller and insignificant. By contrast, street density and amenity density strengthen in Stage 2 relative to Section 6. This pattern is consistent with the hypothesis that walkability indices partly proxy for housing stock quality (walkable cities have newer, larger apartments), while street network density and amenity density more directly measure location quality independent of dwelling characteristics.

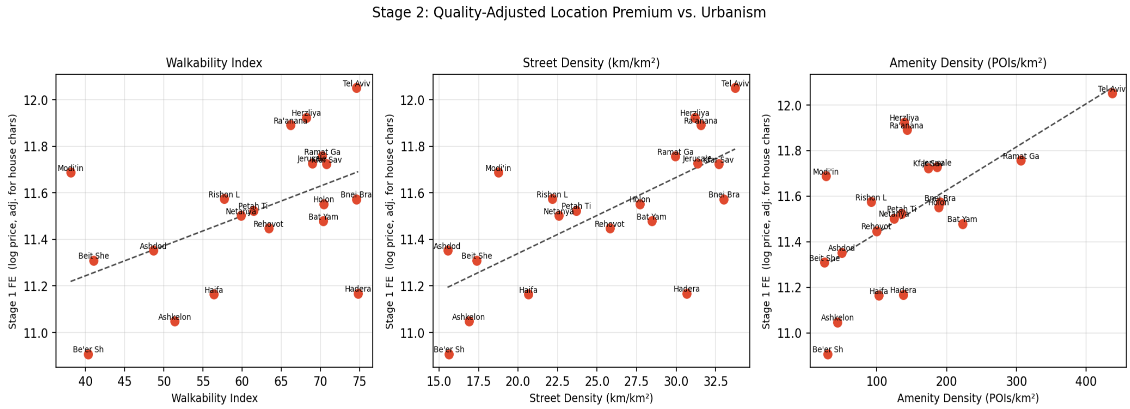

Figure S3. Stage 2 scatter plots: city-aggregated Stage 1 FEs ($\hat{\alpha}$, composition-adjusted) vs. walkability index, street density, and amenity density. Cities at the top-right have both high urbanism quality and high location premiums, controlling for apartment size and age.

10. Conclusion

This paper has linked quantitative urbanism metrics to residential apartment prices across 20 Israeli cities, using administrative transaction data for 2018–2023 and OpenStreetMap network data. Our main empirical findings are that street density and amenity density are strongly positively correlated with city-level median apartment prices (Pearson $r \approx 0.63$–0.64, $p < 0.01$), while junction density ($r = 0.54$) and the composite walkability index ($r = 0.44$) are also significant. Dead-end ratio and circuity are not significantly associated with prices at the city level.

We then connected these empirical findings to the concept of effective housing supply—the quality-adjusted aggregate stock of housing services, analogous to efficiency-weighted labor supply. The observed price premia in walkable, amenity-rich cities correspond precisely to the locational quality multipliers needed to aggregate a heterogeneous housing stock into effective supply units. Calibrating these multipliers for Israel, an apartment in Tel Aviv contributes approximately twice as many effective housing units as a national-median apartment, while an apartment in Be'er Sheva contributes about half as many.

The implications are substantive. Israel's housing shortage is not merely a shortage of physical units; it is a shortage of effective housing services—dwellings in walkable, amenity-rich, well-connected neighborhoods. Policies that add units in low-walkability peripheral cities provide partial relief at best. Effective supply expansion requires either densification of existing walkable neighborhoods or large-scale investment in the street networks and amenities that create walkability.

Several limitations should be noted. City-level correlations do not establish causality; unobserved city characteristics confound the urbanism–price relationship. Our walkability index is constructed from open-source network data without direct measurement of perceived walkability or transit quality. The gush-block level analysis (Section 9) increases the sample to 57,007 observations and confirms all city-level findings with high precision, but inherits city-level urbanism metrics for each block and thus cannot identify within-city effects. Future work should exploit the full cadastral boundary file (TABU sub-gush geometries) to assign sub-city Overture locality urbanism metrics at the gush level, enabling city fixed-effect regressions that cleanly identify within-city walkability and amenity density premia. Ideally, planned infrastructure improvements would be used as instruments—to identify causal effects, and quality-adjusted supply measures embedded directly in Israeli spatial equilibrium models.

Notwithstanding these limitations, the analysis demonstrates that urban form matters for Israeli housing affordability in ways that conventional unit-count approaches miss. As Israel seeks to build its way out of its housing crisis, the quality of where it builds is at least as important as the quantity.

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